

Highlights

The continuum limit of a space-filling layer of cubes: a tiling identity, an exact lattice kernel and the closed-form error of the discrete layer

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- Voxel coupling averaged over the source cell equals the continuum's exactly
- The cube's single-site T-matrix follows from distributional moment integrals
- The voxel scheme's leading error is known in closed form for every grid
- A first-moment voxel converges at fourth order for P, SV and SH waves
- In a random stratified layer, one voxel per model layer reaches 1e-6 accuracy

The continuum limit of a space-filling layer of cubes: a tiling identity, an exact lattice kernel and the closed-form error of the discrete layer

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Abstract

We analyse the convergence of voxel discretisations of the elastic volume integral equation, in which a medium is divided into cubic cells, each represented by a single-site T -matrix and coupled through the background Green's tensor: the elastic counterpart of the discrete dipole approximation. Because cubes tile space, a layer of identical voxels is exactly a homogeneous layer, so it isolates the discretisation error, with no modelling or shape error. We show that (i) the cube's lateral form factor vanishes on the reciprocal lattice, so the source-cell-averaged coupling equals the continuum's; (ii) the single-site T -matrix follows from distributional moments of the Green's tensor and, against exact scattering by a sphere, is exact statically and departs only by a real, closed-form $(ka)^2$ term; (iii) its self term cancels, so the scheme is a collocation of the continuum equation, of second order with leading error $+(kd)^2/8$ in reflection and $-(kd)^2/24$ in transmission, which no scalar correction can remove; (iv) a Galerkin voxel carrying Legendre moments of its internal field to degree p converges at order $2p + 2$, with leading error $-(kd)^4/720$ at $p = 1$, for incident P, SV and SH waves at normal and oblique

incidence, and one plane of it reproduces the thin-layer series in the thickness D through $O(D^{2p+2})$. In a randomly stratified layer the orders are unchanged, and the fourth-order voxel reaches an accuracy of 10^{-6} with one voxel per model layer. On a voxelised sphere the staircase error dominates. Every result is verified by an independent implementation or an exact solution.

Keywords: Elastic wave scattering, Volume integral equation, Foldy–Lax multiple scattering, Discrete dipole approximation, Galerkin method, Convergence analysis

1. Introduction

The problem this paper serves. Seismic waves in the crust are scattered by heterogeneity on every scale. The models that describe it are parameterised in many ways (as layers or blocks, by smooth basis functions, or by interfaces), and a calculation that uses one must first sample it onto a grid. A volume-integral scattering calculation represents the sampled model as cells of constant properties and treats every cell as a scatterer: a voxel carrying a single-site T -matrix and coupled to every other voxel through the Green’s tensor of the background, with the multiple-scattering series summed. Where the medium varies within a cell, that sampling is itself an approximation: §9 shows that it caps any voxel at second order, unless the cell’s contrast is expanded to the degree of its field. For the media such a scheme is built for, there is no exact solution, and that is the difficulty. Its error cannot be measured against the answer. Comparison with another numerical method mixes the errors of both. And a refinement study can mislead, because a voxel scheme can converge smoothly to the wrong limit. §4 shows how that hap-

17 pens with the most natural choice of coupling, the Green's tensor evaluated
18 between voxel centres: it carries a local term whose size per unit volume does
19 not shrink as the voxels do, and unless the single voxel's response is matched
20 to cancel it, refinement converges to the wrong answer without revealing it.

21 *Why a homogeneous layer.* This paper therefore starts from the simplest
22 configuration in which a voxel scheme runs in full (every voxel, every coupling,
23 the whole multiple-scattering series, at normal and at oblique incidence) and
24 the exact answer is nevertheless known: a layer filled with identical voxels.
25 Because cubes tile space, such a layer does not approximate a homogeneous
26 layer; it *is* one. Any difference between its reflection and transmission and
27 those of the continuous layer is therefore error of the discretisation alone, with
28 no modelling error and no shape error behind which to hide. A sphere, the
29 usual test object for volume-integral and discrete-dipole methods [1], cannot
30 give this: its error mixes the discretisation with the staircase approximation
31 of a curved surface by cubes. The layer plays the part of the patch test in
32 finite elements: reproducing a uniform medium correctly is necessary, though
33 not sufficient, for a scheme to be trusted in a heterogeneous one. What the
34 layer establishes about the scheme's two ingredients, the coupling between
35 voxels and the response of a single voxel, are properties of the ingredients,
36 which are the same in any medium: that the averaged coupling is exact, that
37 the self term cancels, and the form of the single-site error. §9 confirms that
38 they carry over to a stratified medium; a laterally heterogeneous one remains
39 to be tested.

40 *What the reader gains.* Four things that a heterogeneous test cannot provide.
41 First, a proof, exact at normal incidence, that the voxel scheme converges

42 to the continuum, and the specific ingredient, a coupling averaged over the
 43 source voxel, without which it converges to the wrong answer. Second, the
 44 leading error in closed form for every discretisation, so that a grid can be
 45 chosen for a required accuracy before the calculation is run. Third, a single-
 46 site T -matrix derived from first principles, with its error stated and checked
 47 against the exact solution for a sphere. Fourth, a voxel that converges at
 48 fourth order, and with further Legendre moments at any even order, also in
 49 a randomly stratified model (§9): for a reflection accuracy of 10^{-6} it allows
 50 voxels 58 times larger in each direction than the standard one (§11).

51 *Why cubic voxels.* Cubes are the natural cell for three reasons. They *tile*
 52 *space*, so a uniform region discretised into cubes is still that uniform region,
 53 whereas spheres or point scatterers can only approximate it and carry a pack-
 54 ing error of their own. Their *regular lattice* makes the interaction between
 55 cells translation invariant, so the coupling is a convolution (Toeplitz between
 56 planes, Bloch-periodic within them), and the whole system is solved by fast
 57 transforms. And a cube small compared with the wavelength has a *single-site*
 58 *response in closed form*, the elastic depolarisation of a cube (§6), so the only
 59 approximation a voxel introduces is the internal field it assumes.

60 *Multiple scattering on a lattice.* Replacing a heterogeneous body by a lattice
 61 of small scatterers that interact through the background Green's function
 62 is the multiple-scattering description of Foldy [2] and Lax [3]. In electro-
 63 magnetics its discretised form is the discrete dipole approximation [DDA;
 64 4, 5], whose accuracy has been studied in detail: each dipole's polarisability
 65 must account for the local field of the surrounding lattice, which the lattice
 66 dispersion relation fixes so that an infinite lattice propagates waves as the

continuum does to a given order in kd [6]; the error of the method converges at second order in the dipole spacing when shape errors are absent [1]; and integrating the Green's tensor over each cell rather than sampling it at the centre improves accuracy, most of all for strong contrasts [7]. The elastic problem adds a structure the scalar and electromagnetic ones lack. A cubic voxel responds to the incident field with a force (through its density contrast) and a moment (through its stiffness contrast), so the state at a voxel has nine components (displacement and strain), and the single site's self-interaction is the elastic depolarisation of a cube, the discrete counterpart of the inclusion tensor of Eshelby [8].

This paper. The approach originates in the two-and-a-half-dimensional formulation of Nestor [9]: a medium divided into cells that interact through the background Green's tensor, each carrying a single-site contrast renormalised by its own near field, with the multiple-scattering series summed. This paper extends it to cubic voxels. §2 sets out the discrete medium and §3 the continuous layer it must reproduce. §4 shows why the point coupling fails, and §5 why the source-averaged coupling succeeds exactly. §6 derives the single-site T -matrix from moment integrals, tests it against the exact sphere, and shows that with it the discrete system is a collocation of the continuum equation. §7 gives that collocation's second-order error in closed form, and §8 the fourth-order voxel, for incident P and S at normal and oblique incidence, and its generalisation to Legendre moments of any degree. §9 repeats the tests on a randomly stratified layer and on a medium that varies within a cell, §10 on a sphere, and §11 discusses what the results imply for voxel schemes in general. Every result is derived symbolically in Mathematica and

checked against an independent Python implementation or an exact solution.

2. Setting

The discrete medium. Space is tiled by cubes of side d (half-side $h = d/2$). Cube i carries a contrast $\Delta\lambda, \Delta\mu, \Delta\rho$ from an isotropic background with Lamé parameters λ, μ , density ρ , P and S speeds α, β and P-wave modulus $M_P = \lambda + 2\mu$. Time dependence is $e^{-i\omega t}$; $k_P = \omega/\alpha$, $k_S = \omega/\beta$. Coordinates are ordered (z, x, y) with z down. The field at a voxel is the nine-component state

$$\psi = (u_z, u_x, u_y, \varepsilon_{zz}, \varepsilon_{xx}, \varepsilon_{yy}, 2\varepsilon_{xy}, 2\varepsilon_{zy}, 2\varepsilon_{zx}), \quad (1)$$

displacement followed by engineering strain. A voxel responds with a force and a moment (stress dipole); the 9×9 single-site T -matrix maps the exciting state to them, and the 9×9 lattice Green's tensor $\mathbf{K}(i, j)$ maps a source voxel's force and moment to the state at voxel i . The Foldy-Lax system is

$$\psi_i = \psi_i^0 + \sum_{j \neq i} \mathbf{K}(i, j) \mathbf{T}_j \psi_j, \quad (2)$$

with ψ^0 the incident state. Strain rows of $\mathbf{K}$ are receiver derivatives of the point Green's tensor $\mathbf{G}$; moment columns are derivatives with respect to the receiver coordinate, symmetrised with a factor $\frac{1}{2}$ on the shear components. In that convention the contrast operator is

$$\Delta = \text{blockdiag}(\omega^2 \Delta\rho \mathbf{I}_3, \Delta\mathcal{C}), \quad (3)$$

where $\Delta\mathcal{C}$ is the 6×6 contrast stiffness with $\Delta\lambda + 2\Delta\mu$ and $\Delta\lambda$ in the normal block and $2\Delta\mu$ (not $\Delta\mu$) on the shear diagonal.

110 *The laterally uniform layer.* A layer of thickness D is filled with n planes
 111 of voxels, $d = D/n$, and illuminated at normal incidence. The field is then
 112 laterally uniform, only the zero-lateral-wavenumber (specular) component of
 113 the lattice coupling acts, and Eq. (2) becomes exactly a chain of n planes
 114 coupled by the specular lattice sums

$$\mathbf{S}(m) = \sum_{\mathbf{R}} \mathbf{G}(\mathbf{R} + md\hat{\mathbf{z}}), \quad (4)$$

115 where the sum runs over the lattice vectors $\mathbf{R} = d(0, p, q)$, p, q integers, that
 116 join the centres of the voxels in one plane (a square lattice of pitch d), and
 117 m is the separation of the two planes in units of d (Fig. 1a,b). The question
 118 is whether, and how fast, this chain converges to the continuous layer as
 119 $n \rightarrow \infty$ at fixed D .

120 *Implementations.* Every result is derived symbolically in Mathematica and
 121 compared with an independent numerical implementation or an exact solu-
 122 tion. “The package” below is the Python implementation of the voxel scheme
 123 (`cubic_scattering`): its cube T -matrix, its lattice kernels and its stratified
 124 propagator, which serve both as the object under test and, where validated,
 125 as a second implementation.

126 3. The continuous layer and its thin-layer expansion

127 At normal incidence a plane force drives a one-dimensional problem per
 128 wave type, with modulus M (M_P for P, μ for S), wavenumber k and plane
 129 Green’s function $g(z) = \frac{i}{2Mk} e^{ik|z|}$. The layer’s reflection and transmission
 130 follow from continuity of displacement and traction at its faces; they agree

131 with the stratified propagator used as the reference in this work to 2.5×10^{-8} ,
 132 P and S (`ContinuumLimit_Reference.wl`).

133 For a layer of thickness D with modulus M_1 and density ρ_1 in a back-
 134 ground M_0, ρ_0 , write $Z_j = \sqrt{M_j \rho_j}$ for the impedances, $\zeta = Z_1/Z_0$ for their
 135 ratio and $k_1 = \omega \sqrt{\rho_1/M_1}$ for the layer's wavenumber. The reflection coeffi-
 136 cient expands as

$$R = \frac{ik_1 D}{2}(\zeta - \zeta^{-1}) + \frac{(k_1 D)^2}{4}(\zeta^{-2} - \zeta^2) - \frac{i(k_1 D)^3}{24}(\zeta - \zeta^{-1})(3\zeta^{-2} + 2 + 3\zeta^2) \\ + \frac{(k_1 D)^4}{48}(\zeta^2 - \zeta^{-2})(3\zeta^2 - 2 + 3\zeta^{-2}) + O(D^5), \quad (5)$$

137 exact in the contrast (`ContinuumLimit_ThinLayer.wl`, which checks each
 138 coefficient symbolically and the truncated series against the exact layer).
 139 The first term is

$$\frac{i\omega D}{2} \sqrt{\frac{M_0}{\rho_0}} \left[\frac{\rho_1 - \rho_0}{M_0} - \rho_0 \left(\frac{1}{M_1} - \frac{1}{M_0} \right) \right] : \quad (6)$$

140 to first order a thin layer is an interface across which the traction jumps by
 141 the excess inertia and the displacement by the excess compliance. At oblique
 142 incidence the same statement holds with the first-order system $\partial_z \mathbf{b} = \mathbf{A} \mathbf{b}$
 143 for the P-SV state $\mathbf{b} = (u_x, u_z, t_{xz}, t_{zz})$: the layer acts to $O(D)$ as the jump
 144 $\mathbf{I} + (\mathbf{A}_1 - \mathbf{A}_0)D$, which reproduces the layer's $O(D)$ reflection coefficients
 145 R_{PP}, R_{PS} to 10^{-12} ; the series truncated at $O(D^k)$ converges at exactly order
 146 k ; and a soft thin layer with shear modulus $\mu_1 = D/\eta$ at fixed compliance η
 147 tends to the linear-slip interface as $D \rightarrow 0$. These expansions are the target
 148 a single plane of voxels must reproduce order by order. A plane of voxels
 149 that carry Legendre moments of their internal field to degree p reproduces
 150 Eq. (5) term by term through $O(D^{2p+2})$ and first differs at D^{2p+3} (§8): the

151 mean-only voxel reproduces the equivalent interface and the D^2 term, the
 152 first-moment voxel all four terms shown.

153 4. The specular sums of the point Green's tensor

154 *Why transform the sum.* The specular sum (4) adds, at a receiver, the field
 155 of every voxel of a plane m planes away. In that form it is a poor object to
 156 study. The Green's tensor decays slowly with distance, so the sum converges
 157 slowly, and it hides the one question that matters here: which part of it is
 158 the continuum, and which part is the discretisation. The Poisson summation
 159 formula [10] answers both at once, and is the standard tool for lattice sums
 160 of wave fields [11]. For a function f on the plane and the square lattice of
 161 pitch d it states

$$\sum_{\mathbf{R}} f(\mathbf{R}) = \frac{1}{d^2} \sum_{\mathbf{g}} \hat{f}(\mathbf{g}), \quad \hat{f}(\mathbf{g}) = \int f(\mathbf{x}) e^{-i\mathbf{g} \cdot \mathbf{x}} d^2x, \quad \mathbf{g} = \frac{2\pi}{d}(p, q), \quad (7)$$

162 with p, q integers: a sum of samples of f on the lattice equals a sum of samples
 163 of its Fourier transform on the reciprocal lattice, whose pitch is $2\pi/d$ (Fig. 2).
 164 Applied to the specular sum,

$$\mathbf{S}(m) = \frac{1}{d^2} \sum_{\mathbf{g}} \hat{\mathbf{G}}(\mathbf{g}; md), \quad (8)$$

165 with $\hat{\mathbf{G}}(\mathbf{g}; z)$ the two-dimensional (lateral) transform of the Green's tensor
 166 between planes a distance z apart. Each term has a physical meaning. Trans-
 167 formed laterally, the field of a plane of sources is a superposition of plane
 168 waves, one for each lateral wavenumber, and the lattice selects the wavenum-
 169 bers $\mathbf{g}$. The order $\mathbf{g} = 0$ is the laterally uniform plane wave that a continuous

170 layer radiates: it is the continuum's term. Every other order has $|\mathbf{g}| \geq 2\pi/d$,
 171 far beyond the wavenumber of the field when the voxels are small compared
 172 with the wavelength, so it is evanescent and decays away from the plane as
 173 $e^{-|\mathbf{g}|md} = e^{-2\pi m\sqrt{p^2+q^2}}$, independently of d . The transform therefore splits
 174 the lattice sum exactly into the continuum and a discretisation correction,
 175 and the correction converges after a few orders.

176 *The local term that survives.* In the strain-from-moment block the evanescent
 177 orders carry two powers of $|\mathbf{g}| \propto 1/d$, so their contribution per voxel volume
 178 does *not* vanish as $d \rightarrow 0$: at $m = 1$ it is eleven times the continuum
 179 term (`ContinuumLimit_SpecularSums.wl`, matched to the package's Ewald
 180 lattice sum to 5×10^{-13}). This surviving term is static. Summed over all
 181 planes in slab order, the static lattice sum of the point kernel is exactly
 182 the depolarisation of a uniform plate ($-1/M_P$ on $\varepsilon_{zz} \leftarrow M_{zz}$ and $-1/(2\mu)$
 183 on the normal shears, per unit volume) plus a cubic-symmetric remainder
 184 (`ContinuumLimit_StaticLocal.wl`, to 2×10^{-10}). The point kernel therefore
 185 reproduces the continuum only if that remainder is cancelled by the voxel's
 186 own self term.

187 5. The tiling identity

188 Replace the point coupling by the *source-cell average* $\langle \mathbf{G} \rangle(\mathbf{r}) = d^{-3} \int_{\text{cube}} \mathbf{G}(\mathbf{r}-$
 189 $\mathbf{y}) d\mathbf{y}$, the field at a voxel centre of a uniform polarisation of the source voxel.
 190 Each Poisson order in Eq. (8) is then multiplied by the cube's form factor

$$F(\mathbf{g}) = \text{sinc}(g_x h) \text{sinc}(g_y h), \quad (9)$$

191 which vanishes at every $\mathbf{g} \neq 0$ of the reciprocal lattice since $\text{sinc}(p\pi) = 0$ for
 192 integer $p \neq 0$. This is the Fourier form of the fact that cubes tile the plane,
 193 $\sum_{\mathbf{R}} \chi(\mathbf{x} - \mathbf{R}) = 1$ for the cube's indicator χ : a partition of unity (Fig. 3).
 194 Two exact statements follow.

195 *Between planes.* For $m \neq 0$ every evanescent order is removed and

$$\mathbf{S}_{\text{avg}}(m) = \frac{1}{d^2} \sum_{W=P,S} \hat{\mathbf{G}}_W(0; md) \text{sinc}(k_W h), \quad (10)$$

196 the continuum's plane-wave term times the vertical form factor. The pack-
 197 age's cell-averaged kernel agrees in all 81 entries to 2.8×10^{-16} (`ContinuumLimit_`
 198 `AveragedSums.wl`).

199 *Within a plane.* The sum over *all* cells of the plane, the voxel's own included,
 200 is the field at the mid-plane of a uniform plate of thickness d (Fig. 1c):

$$d^3 \mathbf{S}_{\text{avg}}(0) + \mathbf{S}_{\text{self}} = d^3 \mathbf{P}_0, \quad \mathbf{S}_{\text{self}} = \int_{\text{cube}} \mathbf{G}(-\mathbf{y}) d\mathbf{y}, \quad (11)$$

201 where $\mathbf{P}_0 = d^{-3} \int_{-h}^h \mathbf{G}_{\text{plate}}(-z') dz'$ is elementary: in the one-dimensional
 202 transform each entry is $c_0 + A/(Mk_z^2 - \rho\omega^2)$, so it averages to $c_0/d + A(e^{ikh} -$
 203 $1)/(dMk^2)$. Its only nonzero entries are $u \leftarrow f$ on the diagonal and the
 204 three normal strain-from-moment entries, whose local weights $-1/M_P$ and
 205 $-1/(2\mu)$ are the plate depolarisation of §4. Statically the identity holds for
 206 all 21 independent components of the strain-from-moment tensor to 4×10^{-9} ,
 207 with every cell integral evaluated analytically (below).

208 *Consequence: an exact kernel with no lattice sum.* At normal incidence the
 209 same-plane block of the cell-averaged Foldy–Lax kernel is

$$\mathbf{S}_{\text{avg}}(0) = \mathbf{P}_0 - \mathbf{S}_{\text{self}}/d^3, \quad (12)$$

210 with no Ewald splitting, near shell or asymptotic tail. The route it replaces
 211 (Ewald summation with a near shell averaged by quadrature and an $O(d^2)$
 212 Taylor tail beyond it) carries a static strain-from-moment error of 9.4×10^{-4}
 213 at its default settings, falling to 2.6×10^{-5} only when both the shell radius
 214 and the quadrature order are raised; Eq. (12) is now the package’s kernel at
 215 normal incidence.

216 *Cell integrals without quadrature.* Write the static (Kelvin) tensor as $G_{bc} =$
 217 $(c_A + c_B) \delta_{bc}/r - c_B \partial_b \partial_c r$ with $c_B = (1/\mu - 1/M_P)/8\pi$ and $c_A = 1/(4\pi\mu) - c_B$.
 218 Every cell integral of its derivatives is a cell integral of derivatives of $1/r$ and
 219 r . One application of the divergence theorem moves it onto faces $x_p = c_p \pm h$,
 220 which never contain the origin, and there the two-dimensional antiderivative
 221 in the face coordinates is continuous, so the face integral is a sum over the
 222 face’s corners. The distributional term at $r = 0$ is carried by the divergence
 223 theorem itself. (A triple antiderivative evaluated at the cube’s corners is
 224 *not* usable for repeated derivatives: its arctangent terms jump across the
 225 coordinate planes through the origin, and differentiating twice there discards
 226 exactly that term.) The self term so obtained reproduces the classical closed
 227 form of the cube’s depolarisation to machine precision.

228 6. The single site: the cube T -matrix from moment integrals

229 Every error of a voxel scheme is made at the single site (§11), so the single
 230 site is derived here rather than imported. The classical route imports it. For
 231 an ellipsoid, Eshelby’s uniformity theorem makes the internal strain under a
 232 uniform load uniform, with a depolarisation tensor in closed form [8, 12, 13];
 233 its dynamic generalisation is known for spheres and ellipsoids [14, 15]. For

234 a cube or a general polyhedron the static field of a uniform eigenstrain is
 235 also in closed form [16, 17, 18, 19], but the strain it produces is not uniform,
 236 so there is no exact uniform-strain inclusion to import. In scattering theory
 237 the same object appears as the self-consistent or effective-field closure of
 238 the volume integral equation [20, 21, 22]: a first-order (Born) scatterer [23]
 239 renormalised by the geometric series of its own near field, the step that turns
 240 a contrast into an effective contrast. Nestor [9] introduced the single site
 241 in this form, summing the series of the cell's self integral, with the singular
 242 near field evaluated over an infinitesimal cell. The moments below evaluate
 243 the same self integral over the finite cube, as a distribution. The elastic T -
 244 matrix of Waterman [24] represents a scatterer in spherical multipoles, the
 245 natural basis for a sphere; a voxel in a lattice needs one compatible with the
 246 lattice, and the moments over the cube provide it. The construction below
 247 makes the approximation of a voxel explicit rather than implicit: it derives
 248 the cube's T -matrix from the moments of the elastodynamic Green's tensor
 249 over the cube, and identifies the uniform-strain closure as one truncation of
 250 an exact hierarchy.

251 *6.1. The closed hierarchy*

252 Let the voxel be $V = [-h, h]^3$ with contrast $\delta c_{ijkl} = \Delta\lambda \delta_{ij}\delta_{kl} + \Delta\mu(\delta_{ik}\delta_{jl} +$
 253 $\delta_{il}\delta_{jk})$ and $\Delta\rho$. Expand the internal displacement about the centre to second
 254 degree and project the Lippmann–Schwinger equation onto 1, x_p and x_px_q .
 255 The unknowns are $u_i(0)$, $\partial_p u_i(0)$ and $\partial_p \partial_q u_i(0)$, and the result is the closed

256 system

$$(\delta_{ij} - \omega^2 \Delta \rho \mathbf{G}_{ij}) u_j + (\mathbf{N}_{in,k}^r \delta c_{nksj} - \frac{1}{2} \omega^2 \Delta \rho \mathbf{K}_{ij}^{rs}) \partial_r \partial_s u_j = u_i^0, \quad (13)$$

$$(\delta_{pr} \delta_{ij} + \omega^2 \Delta \rho \mathbf{N}_{ij,p}^r - \mathbf{M}_{in,pk} \delta c_{nkrj}) \partial_r u_j = \partial_p u_i^0, \quad (14)$$

$$-\omega^2 \Delta \rho \mathbf{M}_{ij,pq} u_j + (\delta_{qr} \delta_{ps} \delta_{ij} + \mathbf{Q}_{in,pqk}^r \delta c_{nksj} - \frac{1}{2} \omega^2 \Delta \rho \mathbf{P}_{ij,pq}^{rs}) \partial_r \partial_s u_j = \partial_p \partial_q u_i^0, \quad (15)$$

257 all fields evaluated at the centre. Its coefficients are six moments of the
258 background Green's tensor $\mathbf{G}$ over the voxel,

$$\begin{aligned} \mathbf{G}_{ij} &= \int_V G_{ij} dV, & \mathbf{K}_{ij}^{rs} &= \int_V G_{ij} x_r x_s dV, & \mathbf{N}_{in,k}^r &= \int_V \partial'_k G_{in} x_r dV, \\ \mathbf{M}_{in,pk} &= \int_V \partial'_p \partial'_k G_{in} dV, & \mathbf{P}_{ij,pq}^{rs} &= \int_V \partial'_q \partial'_p G_{ij} x_r x_s dV, & \mathbf{Q}_{in,pqk}^r &= \int_V \partial'_q \partial'_p \partial'_k G_{in} x_r dV, \end{aligned} \quad (16)$$

259 with derivatives taken with respect to the source point, $\partial' = -\partial$. That sign
260 reaches $\mathbf{N}$ and $\mathbf{Q}$ only, and no symmetry, parity or scaling check can detect
261 it, so it is fixed by convention. The moments are graded by (D, W) , the
262 number of derivatives and the degree of the weight: $\mathbf{G}(0, 0)$, $\mathbf{K}(0, 2)$, $\mathbf{N}(1, 1)$,
263 $\mathbf{M}(2, 0)$, $\mathbf{P}(2, 2)$, $\mathbf{Q}(3, 1)$.

264 6.2. The moments as distributions

265 Write the elastodynamic tensor as

$$G_{in}(\mathbf{r}) = \frac{1}{4\pi\mu} \left[\delta_{in} g_S(r) + \frac{1}{k_S^2} \partial_i \partial_n (g_S(r) - g_P(r)) \right], \quad g_c(r) = \frac{e^{ik_c r}}{r} \quad (17)$$

266 [25, 26]. Expanding g_c in powers of r reduces every moment in Eq. (16) to a
267 finite sum of scalar moments

$$E[m; d_1 \dots d_D; w_1 \dots w_W] = \langle \partial_{d_1} \dots \partial_{d_D} r^m, x_{w_1} \dots x_{w_W} \mathbf{1}_V \rangle. \quad (18)$$

For $m = -1$ and $D \geq 2$ the classical integrand is not absolutely integrable. $\partial^2(1/r)$ decays as r^{-3} , its angular average vanishes, and the value of the integral depends on the shape of the region excluded around the origin. This is the source of the δ -function term of the inclusion problem, which a numerical quadrature misses. Without it the depolarisation is wrong; for a sphere it vanishes altogether (§6.3). It is not added afterwards here. The derivatives are distributional from the start, and each is moved onto the voxel's faces by pairing with the test function $w\mathbf{1}_V$, using $\partial_q \mathbf{1}_V = -n_q \delta_{\partial V}$:

$$\langle \partial_q T, w\mathbf{1}_V \rangle = \oint_{\partial V} n_q w T \, dA - \langle T, (\partial_q w) \mathbf{1}_V \rangle. \quad (19)$$

The surface term is evaluated on the faces, a distance h from the origin, where every derivative of r^m is smooth. The recursion lowers D and W together and ends on convergent integrals. The face integrals are done in closed form by the corner sums of §5. The sharp check is the sum rule

$$\sum_p E[-1; p, p;] = \nabla^2(1/r) \text{ paired with } \mathbf{1}_V = -4\pi, \quad (20)$$

which the face route satisfies exactly and a spherical excision returns as 0. The closed forms contain logarithms of the units $(2 \pm \sqrt{3})^n$ of the Pell equation $x^2 - 3y^2 = 1$. Left unreduced, these cancel catastrophically: for example, $\log(72010600134783751 - 41575339372323900\sqrt{3})$ is a cancellation of 35 digits. They are therefore reduced to $n \log(2 - \sqrt{3})$, which is exact. Because every power of r is paired this way, the dynamic moments keep the even-power terms of the expansion $(k^2 r, r^3, \dots)$ as well as the odd ones. A route that integrates only the polynomial (odd-power) terms of $\mathbf{G}$ misses those terms, at a real $O((kh)^2)$ cost (§6.4).

289 6.3. Parity, and the nine-component block

290 The cube is centrosymmetric. A moment that couples a field quantity of
 291 degree a to one of degree b therefore vanishes unless $a + b$ is even. u , ∂u and
 292 $\partial\partial u$ have degrees 0, 1 and 2, so Eq. (14) contains the first gradient alone: ∂u is
 293 determined by its own block and couples to neither u nor $\partial\partial u$. The leading
 294 scattering by a stiffness contrast is the stress dipole $\int_V \delta c : \nabla u \, dV$, which
 295 needs only ∂u . The leading far field of the stiffness contrast is therefore fixed
 296 by Eq. (14), and *no treatment of the second-gradient block (15) can change*
 297 *it*. What Eq. (15) contributes reaches the far field only through the $O((kh)^2)$
 298 coupling in Eq. (13). That is the order, and the origin (the variation of the
 299 field across the voxel), of the collocation error of §7, which the first-moment
 300 voxel of §8 removes.

301 With $\Delta\rho = 0$, Eq. (14) is the 9×9 operator $\delta_{pr}\delta_{ij} - \mathbf{M}_{in,pk}\delta c_{nkrj}$, built
 302 from $\mathbf{M}$ alone. Under O_h the nine gradient components decompose as $A_{1g} \oplus$
 303 $E_g \oplus T_{1g} \oplus T_{2g}$, and the operator is diagonal in that basis. Its eigenvalues are
 304 the inverse strain concentrations:

$$A_{1g} \text{ (dilatation)} : \quad 1 + \frac{\Delta\lambda + \frac{2}{3}\Delta\mu}{\lambda + 2\mu}, \quad (21)$$

$$T_{2g} \text{ (off-diagonal shear)} : \quad 1 + 2\Delta\mu S_{\text{off}}, \quad S_{\text{off}} = \frac{\pi(\lambda + 2\mu) - \sqrt{3}(\lambda + \mu)}{3\pi\mu(\lambda + 2\mu)}, \quad (22)$$

$$E_g \text{ (diagonal shear)} : \quad 1 + 2\Delta\mu S_{\text{diag}}, \quad S_{\text{diag}} = \frac{3\sqrt{3}(\lambda + \mu) + 2\pi\mu}{6\pi\mu(\lambda + 2\mu)}. \quad (23)$$

305 The density channel is the displacement block of Eq. (13) with the second
 306 gradients dropped, an amplification $(1 - \omega^2\Delta\rho\mathbf{G})^{-1}$ with $\mathbf{G}$ the cube integral
 307 of $\mathbf{G}$.

308 *The cube against the sphere.* For a sphere, Eshelby's tensor gives a dilata-
 309 tional channel $1 + \Delta K / (\lambda + 2\mu)$, with $\Delta K = \Delta\lambda + \frac{2}{3}\Delta\mu$, and a single deviatoric
 310 channel $1 + 2\Delta\mu S_{\text{sph}}$, with $2\mu S_{\text{sph}} = 2(4 - 5\nu)/[15(1 - \nu)]$. The cube's di-
 311 latational channel, Eq. (21), is the sphere's exactly: its eigenvalue involves
 312 $\mathbf{M}$ only through the contraction $\int_V \partial_i \partial_n G_{in} dV$, and the static Navier equa-
 313 tion makes $\partial_i \partial_n G_{in} = -\delta(\mathbf{r})/(\lambda + 2\mu)$, whose integral does not depend on
 314 the shape. The cube's shear channels differ from each other. Their isotropic
 315 average, however, is the sphere's, also exactly:

$$\frac{1}{5}(3S_{\text{off}} + 2S_{\text{diag}}) = S_{\text{sph}}, \quad S_{\text{diag}} - S_{\text{off}} = \frac{(\lambda + \mu)(5\sqrt{3} - 2\pi)}{6\pi\mu(\lambda + 2\mu)} > 0. \quad (24)$$

316 The $\sqrt{3}$ terms cancel in the average. What a cube adds to a sphere at this
 317 order is therefore a pure cubic anisotropy. It splits the deviatoric channel
 318 about the sphere's value, weighted 3 : 2 by the dimensions of T_{2g} and E_g ,
 319 and the diagonal shear is the more depolarised. At the background of §7
 320 ($\nu = 7/32$), $2\mu S_{\text{off}} = 0.4314$, $2\mu S_{\text{diag}} = 0.5929$ and the sphere's value is
 321 0.4960.

322 *Where Eshelby enters.* Equations (21)–(23) follow from the moment integrals
 323 and the group theory alone. No ellipsoidal result is used, and no δ term is
 324 added by hand. What corresponds to Eshelby is the *truncation*. Keeping
 325 ∂u and discarding $\partial\partial u$ is exactly the uniform-internal-strain ansatz, and by
 326 the uniformity theorem that ansatz is exact if and only if the inclusion is
 327 an ellipsoid. For a sphere the construction is therefore exact in the static
 328 limit, and for a cube it is an approximation. The hierarchy (13)–(15) states
 329 that approximation as the omitted blocks, rather than as an unquantified
 330 shape correction. A polynomial expansion about the centre cannot represent

331 the edge and corner concentrations of an isolated cube's internal field at any
 332 finite degree. A voxel in a uniform layer has no such concentrations: its
 333 neighbours have the same properties, so its internal field is the continuum's,
 334 which is smooth. That is why a low-degree internal field suffices in a lattice
 335 (§§7–8) although it is only an approximation for an isolated cube.

336 *Against the exact sphere.* These statements are tested against the exact elas-
 337 tic Mie solution for a sphere, with no Eshelby formula in between. The
 338 concentration of Mie order n is $E_n = a_n/a_n^{\text{Born}}$ (orders 0, 1, 2 are the dilata-
 339 tional, density and shear channels), extrapolated to $ka \rightarrow 0$. The formulation
 340 is applied to a ball of radius a , with its moment $\mathbf{M}$ evaluated by the surface
 341 route of Eq. (19) on $|\mathbf{r}| = a$. Its dilatational and shear channels equal Mie's
 342 to 2×10^{-10} , and the density channel is 1. The cube's closed forms meet
 343 the same numbers. Its dilatational channel is Mie's E_0 to 2×10^{-10} , and its
 344 isotropic shear average, Eq. (24), is the depolarisation read from Mie's E_2
 345 to 10^{-8} . The same moment taken as a principal value, over the ball with
 346 a small sphere excluded around the origin, has zero depolarisation in every
 347 channel. The whole static response of the sphere is the δ term.

348 At finite frequency the closure uses the dynamic ball moments, which are
 349 in closed form. It departs from Mie at second order in ka and first order in
 350 the contrast, and the departure is closed-form too: the ball's moments are
 351 elementary, and Mie's low-frequency series with the contrast kept exact gives

352 E_n to relative order $(ka)^3$. To first order in the contrast,

$$\frac{E_n^{\text{closure}}}{E_n^{\text{Mie}}} - 1 = \begin{cases} -\frac{(k_P a)^2}{10} \frac{\Delta\lambda}{\lambda + 2\mu}, & n = 0, \\ \frac{2(k_S a)^2 + (k_P a)^2}{30} \frac{\Delta\rho}{\rho}, & n = 1, \\ -\frac{3(k_S a)^2 + 2(k_P a)^2 \beta^2 / \alpha^2}{75} \frac{\Delta\mu}{\mu}, & n = 2, \end{cases} \quad (25)$$

353 with no $O(ka)$ term and no $O((ka)^3)$ term. The imaginary $(ka)^3$ terms,
 354 the radiation damping, agree exactly in all three channels. So the closure's
 355 only error at this order is the real $(ka)^2$ term, the uniform-field ansatz's
 356 own. That is the order the hierarchy assigns to the omitted second gradi-
 357 ents (§6.3), and it has the same origin, the variation of the field across the
 358 scatterer, as the collocation error of §7. Eq. (25) is derived symbolically
 359 (`SphereClosureDynamic.wl`). It agrees with the coefficient extracted from
 360 the numerical Mie solve and an independent implementation of the closure
 361 to 5×10^{-4} , which is the size of the next order in the contrast.

362 6.4. The collocation closure

363 The collocation closure takes the internal field of a voxel as uniform, the
 364 truncation of §6.3 for the density and stiffness channels together. The voxel's
 365 own field at its centre is then $\mathbf{S}_{\text{self}} \Delta e$ for internal state e , where $\mathbf{S}_{\text{self}}$ is the
 366 nine-component form of the moments $\mathbf{G}$ and $\mathbf{M}$. By parity it is block-diagonal.
 367 So $e = \psi + \mathbf{S}_{\text{self}} \Delta e$ and

$$\mathbf{T} = d^3 \Delta (\mathbf{I} - \mathbf{S}_{\text{self}} \Delta)^{-1}. \quad (26)$$

368 With the kernel (12) the self term cancels. Substituting $\psi = (\mathbf{I} - \mathbf{S}_{\text{self}}\Delta)e$
 369 into Eq. (2),

$$e_i = \psi_i^0 + d^3 \sum_j \mathbf{K}_{\text{all}}(i, j) \Delta e_j, \quad \mathbf{K}_{\text{all}}(i, i) = \mathbf{P}_0, \quad \mathbf{K}_{\text{all}}(i, j) = \mathbf{S}_{\text{avg}}(i - j), \quad (27)$$

370 an identity checked on the matrices to 4×10^{-16} . Every block of $\mathbf{K}_{\text{all}}$ is an
 371 *exact* cell integral of the one-dimensional continuum kernel (Eqs. (10), (11)),
 372 so Eq. (27) is precisely the collocation of the continuum Lippmann–Schwinger
 373 equation with a strain piecewise constant over voxels. It does not contain
 374 $\mathbf{S}_{\text{self}}$ at all.

375 The package’s cube T -matrix is this closure statically: both blocks agree
 376 with Eq. (26) to 2×10^{-5} at $k_S h = 0.01$, the difference falling as $(kh)^2$
 377 (`ContinuumLimit_CubeT.w1`). It departs dynamically by about $0.25(kh)^2$
 378 through its far-field form-factor corrections. Its self integrals are exact in the
 379 static and radiation parts but omit the real even-power terms of the Green’s
 380 tensor’s expansion ($k^2 r, r^3, \dots$), an error of 0.4–0.6 $(kh)^2$ against the analytic
 381 self term, whose series agrees with the closed-form elastodynamic tensor to
 382 10^{-14} . By Eq. (27) that error cannot enter the closure scheme.

383 7. Convergence, and the closed-form error

384 *Second order.* For a layer $D = 2\text{ m}$ with the contrast $\Delta\lambda = 2\text{ GPa}$, $\Delta\mu =$
 385 1 GPa , $\Delta\rho = 100\text{ kg m}^{-3}$ in a background $\alpha = 5000$, $\beta = 3000\text{ m s}^{-1}$, $\rho =$
 386 2500 kg m^{-3} , illuminated by a P plane force, the reflected displacement of
 387 the chain converges to the continuous layer’s at order 2.000 for $n = 4 \rightarrow$
 388 16 at $\omega = 60, 300$ and 600 rad s^{-1} , with the collocation (27) and with the

389 package *T*-matrix alike (`ContinuumLimit_Chain.wl`). The error depends on
 390 $k_S h$ alone.

391 *The leading error in closed form.* In Eq. (27) the kernel is integrated exactly;
 392 the only approximation is the internal field's variation across a voxel. At first
 393 order in the contrast the discrete scattered field is therefore a midpoint rule
 394 applied to the continuum's Born integral, with the kernel's cell integral exact.
 395 For reflection the integrand carries the two-way phase e^{2ikz} ; for transmission
 396 it is flat. With k the background wavenumber of the wave concerned (k_P for
 397 an incident P wave, k_S for an incident S wave), R the reflection coefficient
 398 and T^{sc} the scattered part of the transmitted field, for *every* n

$$\frac{R_{\text{disc}}}{R} = \frac{\text{sinc}(kh)}{\text{sinc}(2kh)} = 1 + \frac{(kd)^2}{8} + O(d^4), \quad \frac{T_{\text{disc}}^{\text{sc}}}{T^{\text{sc}}} = \text{sinc}(kh) = 1 - \frac{(kd)^2}{24} + O(d^4), \quad (28)$$

399 confirmed for $n = 1$ to 16 to 10^{-4} of the error itself (`ContinuumLimit_`
 400 `ErrorConstant.wl`); for an incident S wave, with $k = k_S$, the same forms
 401 and those of Eq. (31) hold to 4×10^{-4} (`ContinuumLimit_IncidentS.wl`, §8).
 402 At the full contrast the reflection constant is unchanged (0.124 against 1/8);
 403 the transmission constant grows to -0.092 .

404 *Why a uniform-field single site stops at second order.* The two constants have
 405 opposite signs, so no scalar correction of $\mathbf{T}$, no single $O((kh)^2)$ factor, can
 406 cancel both. The package *T*-matrix illustrates it: its form-factor corrections
 407 reduce the reflection error by a factor 4.45 and *increase* the transmission
 408 error by 1.7, at every n . The missing term in Eq. (28) is the first moment
 409 of the internal field over each voxel, which radiates through the derivative of
 410 the kernel. A fourth-order scheme must carry it: a single site with a gradient

channel, and a kernel whose cell integrals include the kernel's derivative.

8. Fourth order and beyond: Legendre moments of the voxel field

Let each voxel carry the mean *and* the first moment of its internal field. On voxel j , with local coordinate $s \in [-h, h]$ and Legendre functions $\phi_b(s) = P_b(s/h)$ ($\phi_0 = 1$, $\phi_1 = s/h$, $\phi_2 = (3s^2/h^2 - 1)/2$, ...), the field is $w = \sum_b c_{b,j} \phi_b$, $w = (u, \varepsilon)$ the displacement and strain of the P channel, and the polarisation is $q = (\omega^2 \Delta \rho u, \Delta M \varepsilon)$ with $\Delta M = \Delta \lambda + 2\Delta \mu$. Testing the continuum equation

$$w(z) = w^0(z) + \int \mathbf{K}(z-z') q(z') dz', \quad \mathbf{K} = \begin{pmatrix} g & g' \\ g' & g'' \end{pmatrix}, \quad g'' = -k^2 g - \delta(z)/M_P, \quad (29)$$

with the same functions on every voxel gives

$$\left(\int_i \phi_a^2 dz \right) c_{a,i} = \int_i \phi_a w^0 dz + \sum_{j,b} \left[\int_i \int_j \phi_a(z) \mathbf{K}(z-z') \phi_b(z') dz dz' \right] \Delta_1 c_{b,j}, \quad (30)$$

$\Delta_1 = \text{diag}(\omega^2 \Delta \rho, \Delta M)$. Between voxels the double integral is separable, a product of Legendre moments of $e^{\pm i k z}$; within a voxel it is split at $z' = z$ and the local term contributes $-\int_i \phi_a \phi_b dz / M_P$. Every entry is closed-form. Keeping only ϕ_0 recovers a second-order scheme; keeping ϕ_0 and ϕ_1 converges at fourth order in both reflection and transmission (`ContinuumLimit_FourthOrder.wl`; Table 1). A single plane of first-moment voxels represents the 2 m layer 6500 times more accurately than a single plane of uniform ones. The collocation column is the scheme of §6 and reproduces the chain of the full 9×9 system to all printed digits.

Table 1: Relative error of the scattered displacement of the layer, reflection / transmission, at $\omega = 300 \text{ rad s}^{-1}$, against the number of voxel planes n : collocation, the mean-only voxel and the first-moment voxel of Eq. (30). The order is 4.00 at $\omega = 60$ and 600 rad s^{-1} too.

n	collocation	mean only	mean and first moment
1	$1.8 \times 10^{-3} / 1.3 \times 10^{-3}$	$1.2 \times 10^{-3} / 1.8 \times 10^{-3}$	$2.8 \times 10^{-7} / 2.8 \times 10^{-7}$
4	$1.1 \times 10^{-4} / 8.3 \times 10^{-5}$	$7.5 \times 10^{-5} / 1.1 \times 10^{-4}$	$1.1 \times 10^{-9} / 1.1 \times 10^{-9}$
16	$7.0 \times 10^{-6} / 5.2 \times 10^{-6}$	$4.7 \times 10^{-6} / 6.9 \times 10^{-6}$	$4.2 \times 10^{-12} / 4.2 \times 10^{-12}$
order	2 / 2	2 / 2	4 / 4

429 *Higher moments.* Carrying the Legendre moments to degree p changes nothing in the construction: the moments and the same-cell double integrals of
430 $P_b(s/h)$ against exponentials are again closed-form, and for $p \leq 1$ the general-
431 degree solver reproduces the chain above exactly. The second-moment voxel
432 ($p = 2$) converges at sixth order and the third-moment voxel ($p = 3$) at
433 eighth, in reflection and transmission (Table 2; `ContinuumLimit_SecondMoment`.
434 `w1`); the second-moment voxel's order is 6.00 at $\omega = 60, 300$ and 600 rad s^{-1} .
435 The errors fall to 10^{-22} , so they are computed in 40-digit arithmetic.

437 *The leading error in closed form.* Without scattering, the Galerkin solution
438 is the L^2 projection Π of the incident field onto each voxel's basis, and the
439 kernel is integrated over each voxel exactly; at first order in the contrast the
440 error is therefore the kernel's cell integral applied to $\Pi w^0 - w^0$. The field
441 varies as e^{iks} across a voxel and the kernel toward an outside observer as e^{iqs} ,
442 $q = +k$ for reflection and $q = -k$ for transmission, so for every scheme and

Table 2: Relative error of the scattered displacement of the layer, reflection / transmission, at $\omega = 300 \text{ rad s}^{-1}$, for voxels carrying Legendre moments to degree p . The order is the mean over the halvings $n = 2 \rightarrow 4 \rightarrow 8$.

n	first moment ($p = 1$)	second moment ($p = 2$)	third moment ($p = 3$)
1	$2.8 \times 10^{-7} / 2.8 \times 10^{-7}$	$2.8 \times 10^{-11} / 4.2 \times 10^{-11}$	$1.5 \times 10^{-15} / 1.5 \times 10^{-15}$
2	$1.7 \times 10^{-8} / 1.7 \times 10^{-8}$	$4.4 \times 10^{-13} / 6.5 \times 10^{-13}$	$6.0 \times 10^{-18} / 6.0 \times 10^{-18}$
4	$1.1 \times 10^{-9} / 1.1 \times 10^{-9}$	$6.9 \times 10^{-15} / 1.0 \times 10^{-14}$	$2.4 \times 10^{-20} / 2.4 \times 10^{-20}$
8	$6.7 \times 10^{-11} / 6.7 \times 10^{-11}$	$1.1 \times 10^{-16} / 1.6 \times 10^{-16}$	$9.2 \times 10^{-23} / 9.2 \times 10^{-23}$
order	4 / 4	6 / 6	8 / 8

every n

$$\frac{\text{discrete}}{\text{exact}} = \frac{\int_{-h}^h e^{iqs} \Pi[e^{iks}] ds}{\int_{-h}^h e^{i(q+k)s} ds}, \quad (31)$$

with Π the centre value for collocation, the cell mean for the mean-only voxel and the projection onto $\{P_0, \dots, P_p\}$ for the voxel of degree p (`ContinuumLimit_FourthOrderError.wl`, `ContinuumLimit_SecondMoment.wl`). The results are in Table 3. The chain of Eq. (30) reproduces the forms for collocation and degrees 0 and 1 at $n = 1$ to 8 to 10^{-4} of the error, the remainder being the second order in the contrast, and those for degrees 2 and 3 to 4×10^{-7} at a contrast reduced by 10^{-6} . The constant for degree p follows from the Legendre expansion $e^{iks} = \sum_j i^j (2j+1) j_j(kh) P_j(s/h)$. The projection drops the terms $j > p$, led by $j = p+1$, whose coefficient is $O((kh)^{p+1})$; its moment against e^{iqs} is $2h i^{p+1} j_{p+1}(qh) = O((qh)^{p+1})$. With $j_n(x) \approx x^n / (2n+1)!!$ and

Table 3: The leading error of the discrete layer in closed form, Eq. (31): discrete over exact scattered field at first order in the contrast, for every number of voxel planes n .

	reflection	transmission
collocation	$1 + (kd)^2/8 + O(d^4)$	$1 - (kd)^2/24 + O(d^4)$
mean only	$1 + (kd)^2/12 + O(d^4)$	$1 - (kd)^2/12 + O(d^4)$
first moment	$1 - (kd)^4/720 + O(d^6)$	$1 - (kd)^4/720 + O(d^6)$
second moment	$1 + (kd)^6/100800 + O(d^8)$	$1 - (kd)^6/100800 + O(d^8)$
third moment	$1 - (kd)^8/25401600 + O(d^{10})$	$1 - (kd)^8/25401600 + O(d^{10})$
degree p	$1 + (-1)^p c_p (kd)^{2p+2}$	$1 - c_p (kd)^{2p+2}$

454 $h = d/2$,

$$c_p = \frac{2p+3}{4^{p+1}[(2p+3)!!]^2} = \frac{[(p+1)!]^2}{(2p+2)!(2p+3)!}, \quad c_0 = \frac{1}{12}, \quad c_1 = \frac{1}{720}, \quad c_2 = \frac{1}{100800}, \quad (32)$$

455 and the leading error carries $(q/k)^{p+1}$: the same sign in reflection and trans-
456 mission for odd p , opposite signs for even p . The first-moment voxel's leading
457 error therefore does not depend on the direction in which the field is observed.
458 Eq. (32) is checked symbolically for $p = 0$ to 5 (every lower power of kd van-
459 ishes) in Mathematica and, independently, in sympy.

460 *One plane of voxels and the thin-layer series.* Refining at fixed thickness is
461 one limit; thinning a single plane is the other. For one plane of voxels ($n = 1$,
462 $d = D$) the series in D of the reflection coefficient, computed exactly at the
463 full contrast, coincides with that of the exact layer, Eq. (5), term by term
464 through D^{2p+2} and first differs at D^{2p+3} , for $p = 0$ to 3 and for a layer both
465 stiffer and softer than its surroundings (`ContinuumLimit_SecondMoment`).

466 w1). Each further Legendre moment adds two terms of the thin-layer series.

467 *In three dimensions.* The three-dimensional first-moment voxel carries the
 468 nine-component state with Legendre weights $\{1, x/h, y/h, z/h\}$, 36 unknowns
 469 in all, tested with the same functions and coupled by cell-to-cell double in-
 470 tegrals of the Green's tensor. For a laterally uniform layer at normal in-
 471 cidence it reduces *exactly* to Eq. (30) (ContinuumLimit_Reduction3D.w1).
 472 The lateral first moments are never excited: the lattice and the incident
 473 field are symmetric under reflection about every voxel's centre planes, and
 474 $x/h, y/h$ are odd there. The mean and the z -moment carry weights that
 475 are constant laterally, so their lateral form factor $\text{sinc}(g_x h) \text{sinc}(g_y h)$ van-
 476 ishes on the reciprocal lattice and the plane sum of the double integrals
 477 is the plate-to-plate integral in z alone. Two facts mark where genuinely
 478 three-dimensional structure begins. The lateral first moment's form fac-
 479 tor, $F_1(g) = i(gh \cos gh - \sin gh)/(gh)^2$, does not vanish on the recipro-
 480 cal lattice ($F_1 = i(-1)^p/p\pi$), so the gradient channels couple through the
 481 evanescent orders. And at oblique incidence the mean's own zeros shift,
 482 $\text{sinc}((2\pi p/d + k_x)h) = (-1)^p k_x h/p\pi + O(k_x^2 h^2)$, so the tiling identity holds
 483 only to $O(k_{\parallel} h)$.

484 *Oblique incidence.* The scheme is built spectrally. By Poisson summation
 485 the Bloch coupling between planes i and j at lateral wavenumber $\mathbf{k}_{\parallel}$ is

$$C_{ab}(i, j) = \frac{1}{d^2} \sum_{\mathbf{g}} L_a(\boldsymbol{\kappa}) L_b(-\boldsymbol{\kappa}) \iint \phi_a(z) \hat{\mathbf{G}}(\boldsymbol{\kappa}; z-z') \phi_b(z') dz dz', \quad \boldsymbol{\kappa} = \mathbf{k}_{\parallel} + \mathbf{g},$$

(33)

486 with L_a the lateral Legendre factors. The whole-space tensor decays as
 487 $1/k_z^2$ and the nine-component state adds at most two powers of k_z , so each

entry of $\hat{\mathbf{G}}(\boldsymbol{\kappa}; z)$ is a $\delta(z)$ of constant weight plus, per mode, $[\mathrm{i}\alpha/2\gamma + \mathrm{i}\beta \operatorname{sgn}(z)/2] \mathrm{e}^{\mathrm{i}\gamma|z|}$, $\gamma^2 = k_W^2 - \kappa^2$: every z integral is closed-form. Double averaging makes the lattice sum converge without Ewald splitting, to three digits at $|p|, |q| \leq 8$. Against the exact layer, for a P wave incident at 20° and 40° (`ContinuumLimit_Oblique.wl`; Table 4). The mean-only voxel is

Table 4: Relative error of the reflected P (R_{PP}) and converted S (R_{PS}) waves for a P wave incident at 20° , with the mean-only and first-moment voxels of Eq. (33). At 40° the orders are again 2 and 4.

	mean only		mean and first moments	
n	R_{PP}	R_{PS}	R_{PP}	R_{PS}
1	9.3×10^{-4}	1.7×10^{-3}	7.2×10^{-8}	4.1×10^{-7}
2	2.3×10^{-4}	4.2×10^{-4}	4.5×10^{-9}	2.5×10^{-8}
8	1.5×10^{-5}	2.7×10^{-5}	1.7×10^{-11}	9.9×10^{-11}
order	2	2	4	4

second order and the first-moment voxel fourth order in both the reflected P and the converted S wave: the breakdown of the tiling identity at $O(k_{\parallel}h)$ and the evanescent coupling of the gradient channels do not cost an order. At $\mathbf{k}_{\parallel} = 0$ the construction reproduces Eq. (30) to all printed digits.

Incident S. Nothing in the construction is specific to P. For a normally incident S wave the layer is the one-dimensional problem of Eq. (30) with $M_P \rightarrow \mu$, $k_P \rightarrow k_S$ and $\Delta M \rightarrow \Delta\mu$; the collocation and mean-only schemes converge at order 2.00 and the first-moment voxel at 4.00, in reflection and transmission, and the full 9×9 construction with an incident SH wave reproduces that one-dimensional result to all printed digits. At oblique inci-

503 dence, against the exact layer (`ContinuumLimit_IncidentS.wl`), the orders
are those of Table 5. The orders, and with $k = k_S$ the closed-form constants

Table 5: Observed order of convergence ($n = 2 \rightarrow 8$, $\omega = 300 \text{ rad s}^{-1}$) for incident SV and SH waves, with the mean-only and first-moment voxels. The SV angles lie below the critical angle $\arcsin(\beta/\alpha) = 36.9^\circ$.

	mean only	mean and first moments
SV at 20° : R_{SS} / R_{SP}	2.00 / 2.00	4.00 / 3.99
SV at 30° : R_{SS} / R_{SP}	1.98 / 2.00	3.97 / 4.00
SH at 20° and 40° : R_{SH}	2.00	4.00

504

505 of Eqs. (28) and (31), are those of the incident P wave.

506 The spectral construction is implemented twice. The second implementa-
507 tion (`scripts/crosscheck_first_moment_voxel.py`) shares only the defi-
508 nition of the scheme: it takes the depth dependence of the transformed kernel
509 from an independent residue construction, obtains the local weight as a nu-
510 merical large- k_z limit of the inverted Christoffel matrix, evaluates every dou-
511 ble integral by quadrature and separates P from S by sampling the upgoing
512 field at two depths. The raw reflected displacements of the two agree to $9 \times$
513 10^{-16} in all 36 cases compared: incident P, SV and SH, normal incidence and
514 20° , $n = 1, 2, 4$, with and without the first moments. The one-dimensional
515 scheme of any degree is also implemented twice. The second implementation
516 (`scripts/crosscheck_second_moment_voxel.py`) evaluates every integral
517 by Gauss quadrature, split at the receiver within a voxel, and the exact layer
518 from its own interface system, in double precision. Its errors agree with the
519 40-digit ones within its round-off floor (3×10^{-14} in reflection, 1.6×10^{-13} in

520 transmission); at $\omega = 1200 \text{ rad s}^{-1}$ it measures orders 4.02 / 3.99 and 6.02 /
 521 6.00; for one plane as the layer thins its differences from the exact layer fall
 522 as $D^{3.00}$, $D^{5.00}$ and $D^{7.00}$ for $p = 0, 1$ and 2 ; and it rederives c_p for $p \leq 5$.
 523 Figure 4(a,b) shows the convergence, with the orders fitted to it.

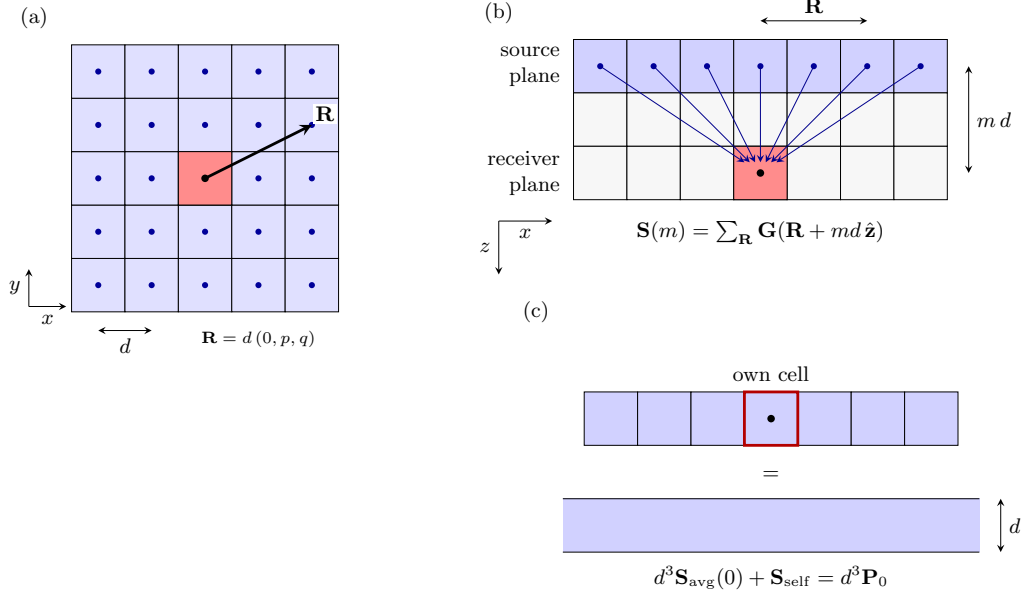


Figure 1: The lattice and its sums. (a) Plan view of one plane of voxels: their centres form a square lattice of pitch d , and a lattice vector $\mathbf{R} = d(0, p, q)$ joins the centre of one voxel (red) to another. (b) Cross-section (z down), between planes: the voxel of a receiver plane (red) is coupled to every voxel of a source plane m planes above it, one for each lattice vector $\mathbf{R}$; the sum over $\mathbf{R}$ is the specular lattice sum of Eq. (4). (c) Within a plane: the sum over every cell of the plane, the receiver’s own cell (outlined) included, is the field of a uniform plate of thickness d , the tiling identity of Eq. (11).

Alt text: Three schematic panels. On the left, a plan view of a five-by-five grid of square voxels with a dot at each centre; the centre voxel is highlighted, an arrow labelled $\mathbf{R}$ runs from its centre to another voxel centre, and the spacing between neighbouring centres is marked d . At the top right, a cross-section through three rows of voxels: arrows from the centre of every voxel in the top row converge on a highlighted voxel two rows below, with the plane separation $m d$ marked. Below it, a single row of voxels with one outlined as the receiver’s own cell is shown equal to a continuous shaded band of the same thickness d , a uniform plate.

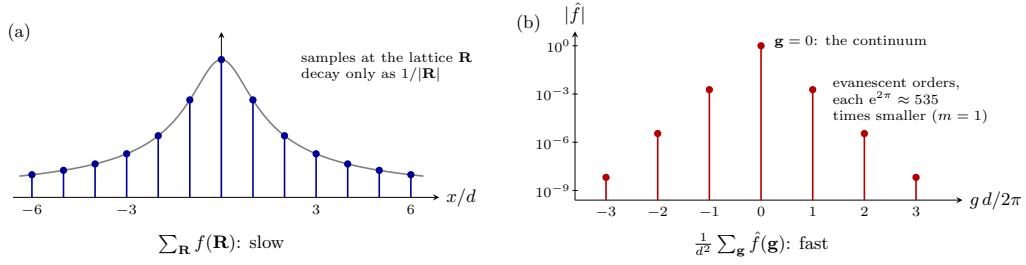


Figure 2: The Poisson summation formula (7) applied to a plane of voxels, along one lateral axis. (a) Real space: the field of each voxel of a source plane at a receiver one plane away, sampled at the lattice points; the samples decay only as $1/|\mathbf{R}|$, so their sum converges slowly. (b) The same sum after Poisson summation, on a logarithmic scale: samples of the lateral Fourier transform at the reciprocal-lattice orders $\mathbf{g} = 2\pi p/d$. The order $\mathbf{g} = 0$ is the continuum's plane wave; every other order is evanescent and, at $m = 1$, smaller than its neighbour by $e^{2\pi} \approx 535$. Magnitudes are schematic: the static decay $e^{-|\mathbf{g}|md}$ is shown.

Alt text: Two panels. On the left, a smooth bell-shaped curve that decays slowly on both sides is sampled by vertical stems at integer positions from minus six to six, the stems remaining clearly visible far from the centre. On the right, on a logarithmic vertical axis from one down to ten to the minus nine, a stem of height one at zero is labelled as the continuum, and stems at plus and minus one, two and three fall by about three powers of ten per step and are labelled as evanescent orders.

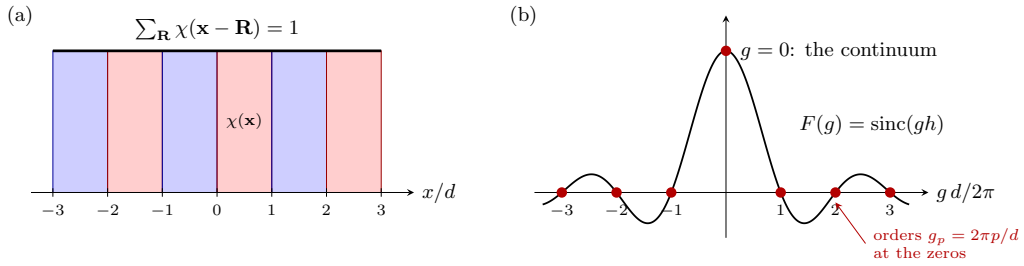


Figure 3: Why cubes make the coupling exact. (a) Cubes tile the plane: the indicator functions χ of the translated cells sum to exactly one everywhere, a partition of unity. (b) The same fact in Fourier space: the cube's lateral form factor $F(g) = \text{sinc}(gh)$ (shown along one axis) vanishes at every reciprocal-lattice order $g_p = 2\pi p/d$ with $p \neq 0$, so of all the orders in the lattice sum (8) only $g = 0$, the continuum's plane wave, survives the source-cell average.

Alt text: Two panels. On the left, adjacent box functions of unit height and unit width, alternately shaded, lie side by side along an axis marked in voxel widths, and a thick horizontal line at height one marks their sum. On the right, the sinc curve, equal to one at the origin and oscillating with decaying amplitude, is plotted against wavenumber in units of $2\pi/d$; dots mark the reciprocal-lattice orders, one at height one at the origin and all the others on the axis at the zeros of the curve.

524 9. A heterogeneous layer

525 A heterogeneous medium is not a layer of identical cells. The simplest
526 heterogeneous case with an exact answer is a stratified model built from layers
527 of constant properties, with real interfaces between them: the form of many
528 layered crustal velocity models. The last test takes such a model. A layer
529 $D = 4$ m thick is divided into eight model layers of 0.5 m, and in each one $\Delta\lambda$,
530 $\Delta\mu$ and $\Delta\rho$ are drawn uniformly within $\pm 15\%$ of the background (the values
531 are listed in `ContinuumLimit_Heterogeneous.w1`). The result is a random
532 stratification with physical contrasts. The voxels refine the model's layering,
533 with m planes of voxels in each model layer, so no voxel straddles a model
534 interface: the natural discretisation when the model's interfaces are known.
535 At $\omega = 1500$ rad s⁻¹ the layer is $k_P D = 1.2$ and $k_S D = 2.0$ thick, so the field
536 reverberates between the model interfaces. The exact answer is the product
537 of the model layers' propagators. Its same-type reflection coefficients agree
538 with those of the reflection–transmission recursion of Kennett [27], computed
539 independently, to 10^{-14} . The schemes are those of §§7–8, with the contrast
540 now changing from plane to plane; nothing else changes. The errors are in
541 Table 6 and Fig. 4(c).

542 *The interfaces cost nothing.* Every order is the one measured for the uniform
543 layer. At a model interface the strain of the true field jumps. A voxel basis
544 that is discontinuous from voxel to voxel represents that jump exactly when
545 the interface lies on a voxel boundary, so the interface adds no error of its
546 own. The only requirement is that voxel boundaries coincide with the model's
547 interfaces. A voxel that straddled an interface would carry a contrast that is
548 not uniform within it, a case not examined here.

Table 6: Relative error against the exact stratified layer, for m planes of voxels per model layer (§9). At normal incidence the order is measured from $m = 4$ to 16, where it is also 2.00 for collocation and 4.00 in transmission; at 20° it is measured from $m = 2$ to 4.

		$m = 1$	$m = 2$	$m = 4$	order
P, normal, R	mean only	1.9×10^{-3}	4.7×10^{-4}	1.2×10^{-4}	2.00
	first moment	8.5×10^{-7}	5.3×10^{-8}	3.3×10^{-9}	4.00
S, normal, R	mean only	5.3×10^{-3}	1.3×10^{-3}	3.3×10^{-4}	2.00
	first moment	6.2×10^{-6}	3.9×10^{-7}	2.4×10^{-8}	4.00
P, 20° , R_{PP}	mean only	1.4×10^{-3}	3.6×10^{-4}	9.0×10^{-5}	2.00
	first moments	3.3×10^{-7}	2.1×10^{-8}	1.3×10^{-9}	4.00
P, 20° , R_{PS}	mean only	2.8×10^{-3}	6.9×10^{-4}	1.7×10^{-4}	2.00
	first moments	1.4×10^{-6}	8.8×10^{-8}	5.5×10^{-9}	4.00

549 *What fourth order buys.* For a reflection error of 10^{-6} at normal incidence
550 (P), the mean-only voxel needs 347 planes of voxels, about 43 per model
551 layer. The first-moment voxel needs 8, one per model layer: the model's own
552 layering, which already gives 8.5×10^{-7} . Counting unknowns (9 per voxel
553 against 36), that is 3123 against 288 per vertical column of voxels, a factor
554 of eleven. In three dimensions, with the lateral grid refined as the vertical
555 one is, the number of voxels falls by $(347/8)^3 \approx 8 \times 10^4$ and the number of
556 unknowns by about 2×10^4 . The same model has been solved independently
557 in Python, with a contrast for each plane: the raw reflected displacements
558 agree with the Mathematica scheme to 1.1×10^{-15} at normal incidence and
559 at 20° , with and without the first moments. Lateral heterogeneity needs the
560 full three-dimensional solver; it is the first extension discussed below.

561 *A medium that varies within a cell.* The stratified model is exact on its own
 562 grid: every cell is homogeneous. A smoothly varying medium is not, and
 563 sampling it as cells of constant properties is an approximation of its own.
 564 Let the contrast of cell j be its Legendre projection to degree r , $\Delta(z) =$
 565 $\Delta_0 \sum_{c \leq r} f_c^{(j)} P_c(s/h)$, with the voxel's field of degree p as in §8. The product
 566 of contrast and field is again a Legendre sum, $P_c P_b = \sum_e A_{cbe} P_e$, with $A_{cbe} =$
 567 $(2e+1) \begin{pmatrix} c & b & e \\ 0 & 0 & 0 \end{pmatrix}^2$ the square of a Wigner $3j$ symbol, so the coupling of Eq. (30)
 568 becomes

$$\iint_j \phi_a(z) \mathbf{K}(z-z') \Delta(z') \phi_b(z') dz dz' = \sum_{c \leq r} \sum_{e \leq p+r} f_c^{(j)} A_{cbe} \left[\iint_j \phi_a(z) \mathbf{K}(z-z') \phi_e(z') dz dz' \right] \Delta_0 : \quad (34)$$

569 the same kernel moments, now to degree $p+r$, and nothing else new. Against
 570 the exact graded layer, a high-precision solution of $(M(z)u')' + \omega^2 \rho(z)u = 0$,
 571 the order of convergence is

$$\min(2p+2, 2r+2), \quad (35)$$

572 or $2p+2$ when degree r represents the profile exactly (Table 7; `ContinuumLimit_`
 573 `GradedContrast.wl`). A contrast that is constant in each cell therefore
 574 caps every voxel at second order, however rich its field basis, and for the
 575 smooth profile the voxel of degree 2 with a linear contrast has the error
 576 of the voxel of degree 1 to within 0.06%. At first order in the contrast
 577 this is what one expects: the contrast's projection error is orthogonal to
 578 the polynomials of degree r , so against the smooth product of the inci-
 579 dent and adjoint fields it contributes at $O(h^{2r+2})$. To keep order $2p+2$
 580 the contrast must be represented to the degree of the field. The closed forms
 581 of the construction are tested one by one by an independent implementa-

Table 7: A contrast that varies within the cells: the field of each voxel to Legendre degree p , its contrast to degree r . Relative reflection error at $n = 16$ planes and the order ($n = 4 \rightarrow 8 \rightarrow 16$), $\omega = 300 \text{ rad s}^{-1}$, for the profiles $1 + (2z/D - 1)/2$ (linear) and $1 + \frac{1}{2} \sin(2\pi z/D)$ (smooth) times the contrast of §7. Predicted: Eq. (35), and $2p + 2$ for the linear profile once $r \geq 1$.

p	r	linear profile		smooth profile	
		error	order	error	order
0	0	7.7×10^{-5}	2.00		
1	0	7.6×10^{-5}	2.00	5.5×10^{-5}	1.95
1	1	6.4×10^{-11}	4.00	1.4×10^{-7}	3.95
2	1	2.3×10^{-17}	6.00	1.4×10^{-7}	3.95
2	2			1.6×10^{-10}	5.98

tion (`scripts/crosscheck_graded_contrast.py`): the Legendre moments
of the kernel against $2h i^n j_n(ch)$ and 30-digit quadrature (to 2×10^{-16}), the
linearisation coefficients against the $3j$ symbols (exactly), and the same-cell
double integrals against 30-digit quadrature (1.4×10^{-16}). It then assem-
bles the scheme without linearisation, evaluating the projected contrast at
the quadrature nodes, against its own 30-digit solution of the graded layer;
its errors agree with those of Table 7 to 10^{-3} , and at $\omega = 1200 \text{ rad s}^{-1}$ it
measures the orders of Eq. (35) again.

10. A sphere: what the layer predicts in three dimensions

The layer and the stratified layer are laterally uniform. A sphere is not,
and elastic scattering by a sphere, homogeneous [28, 29] or radially varying
[30], has an exact solution: it is the usual benchmark for volume-integral and

594 discrete-dipole methods. This section solves two spheres with the full three-
 595 dimensional voxel scheme. The sphere is voxelised by the cubes of an n_{sub}^3
 596 grid whose centres lie inside it, and the Foldy–Lax system is solved with the
 597 uniform-field voxel (the second-order scheme of §§6–7; the first-moment voxel
 598 of §8 has not been built for finite three-dimensional scatterers) and the cell-
 599 averaged coupling, by an FFT-accelerated iterative solver. The single site
 600 is the package’s cube T -matrix, which differs from the collocation closure
 601 only through its far-field corrections (§6.4) and has the same order. A P
 602 wave is incident. The far field is computed from the solution for the incident
 603 plane wave itself, each voxel’s displacement and strain carrying the plane-
 604 wave phase at its centre; adding to that phase a linear variation about the
 605 scatterer’s centre, as a Taylor expansion of the incident field would, counts the
 606 wave’s variation twice and leaves an error of order $(ka)^2$ that no refinement
 607 removes. The error is the largest difference between the scattered far field
 608 and the exact one over nine scattering angles from 11° to 169° , relative to
 609 the largest exact value, at a distance of 5×10^4 radii where the far field has
 610 converged.

611 *A sphere without a staircase.* A voxelised sphere with a sharp surface changes
 612 its shape with every grid. A sphere whose contrast falls smoothly to zero at its
 613 surface does not: the voxels that cross the surface carry a vanishing contrast.
 614 The body here has a homogeneous core $r < a/2$ carrying the contrast of §7,
 615 and across the shell $a/2 < r < a$ the contrast falls to zero as the smoothstep
 616 $s(x) = 10x^3 - 15x^4 + 6x^5$, $x = (a - r)/(a - a/2)$, which vanishes like $(a - r)^3$
 617 at the surface. Its exact scattering follows, order by order, from the radial
 618 equations of Takeuchi and Saito [30], integrated across the shell from the

core's regular solutions and matched to outgoing waves at $r = a$. The equations are derived symbolically from the equations of motion, and the solution is checked against a sphere of homogeneous concentric shells in closed form, by unitarity and reciprocity, and by an independent implementation that agrees to 2×10^{-11} (TakeuchiSaito.wl, ContinuumLimit_GradedSphere.wl, scripts/crosscheck_graded_sphere.py). Each voxel carries the contrast of its centre. Table 8 shows that the error now falls steadily, from 0.47% at $n_{\text{sub}} = 6$ to 0.061% at $n_{\text{sub}} = 12$ at $k_{\text{Sa}} = 0.5$, and from 0.84% to 0.19% at $k_{\text{Sa}} = 1.0$ (scripts/pilot_graded_sphere_vs_exact.py). The apparent order is between 1.7 and 3.6: the second order of the uniform-field voxel with the error of sampling the profile at the voxel centres superposed, which converges irregularly on a cubic lattice. At $n_{\text{sub}} = 4$ a voxel is as wide as the graded shell and the error is 10 to 12%.

Table 8: The smoothly graded sphere, voxelised on grids of n_{sub} voxels across, against its exact solution: relative far-field error, and the apparent order between successive grids.

n_{sub}	voxels	$k_{\text{Sa}} = 0.5$		$k_{\text{Sa}} = 1.0$	
		error	order	error	order
4	32	1.02×10^{-1}		1.15×10^{-1}	
6	136	4.66×10^{-3}	7.62	8.43×10^{-3}	6.45
8	280	2.58×10^{-3}	2.06	5.17×10^{-3}	1.70
12	912	6.11×10^{-4}	3.55	1.88×10^{-3}	2.49

A sphere with a sharp surface. The homogeneous sphere is the familiar test, and its voxelisation changes with every grid. Table 9 shows its error for grids of 4, 6 and 8 voxels across: 4.6%, then 19%, then 4.2% at $k_{\text{Sa}} = 0.5$. The

error tracks the staircase's volume, which is 5%, 17% and 4% too large on these grids. Rescaled by the ratio of the sphere's volume to the staircase's, it falls below 1% at $k_S a = 0.5$ and to at most 3% at $k_S a = 1.0$, but it still does not converge: what remains is the staircase's shape. A refinement study of a sharp sphere measures its staircase; the smooth sphere, like the layer, measures the scheme.

Table 9: A homogeneous sphere voxelised on grids of n_{sub} voxels across, against exact elastic scattering: relative far-field error, raw and rescaled by the ratio of the sphere's volume to the staircase's. It does not converge with refinement, because the staircase changes with the grid.

n_{sub}	voxels	$k_S a = 0.5$		$k_S a = 1.0$	
		raw	volume-rescaled	raw	volume-rescaled
4	32	4.6×10^{-2}	1.4×10^{-3}	5.0×10^{-2}	5.4×10^{-3}
6	136	1.94×10^{-1}	7.9×10^{-3}	1.68×10^{-1}	3.0×10^{-2}
8	280	4.2×10^{-2}	2.3×10^{-3}	3.5×10^{-2}	8.9×10^{-3}

11. Discussion

Figure 4 collects the convergence of every test, with the order fitted to each by least squares (`scripts/plot_convergence_orders.py`).

Where the error of a voxel scheme lives. The results separate the two approximations a voxel scheme makes. The *coupling* between voxels makes none: averaged over the source cell it is the continuum's cell integral exactly, between planes by the form factor's zeros and within a plane by the tiling identity, and the self term it omits is exactly the one the single site

649 carries. Every error therefore comes from the *single site*, from the internal
 650 field it assumes. For a uniform field that error is the midpoint rule's, known
 651 in closed form; for a linear field it is the next term of the same expansion.
 652 The converse is the practical warning. A kernel that is not the cell integral
 653 (the point Green's tensor, or a cell average that does not match the single
 654 site's closure) leaves a local term that does not vanish as the voxels shrink
 655 (§4): the evanescent lattice orders carry a static contribution of fixed size per
 656 voxel volume. A scheme built that way converges to the wrong limit, and
 657 refinement cannot reveal it, because the error is scale-invariant.

658 *What the single site is.* The moment hierarchy of §6 turns the uniform-field
 659 single site from a modelling choice into a derived object with a stated error.
 660 Its static part is exact for a sphere, and for a cube it differs from the sphere's
 661 only by the cubic split of Eq. (24). Its dynamic part departs from exact
 662 scattering by a sphere only through the real $(ka)^2$ term of Eq. (25); the
 663 radiation damping is exact. That term and the collocation error of Eq. (28)
 664 have the same origin, the variation of the field across the scatterer, and the
 665 same remedy, a richer internal field. In a lattice the remedy is cheap: the
 666 field a voxel must represent is the smooth continuum field, not the field of
 667 an isolated cube, with its concentrations at the edges and corners.

668 *Relation to the discrete dipole approximation.* The same partition appears in
 669 the electromagnetic literature as the choice of polarisability. The Clausius–
 670 Mossotti relation, the lattice dispersion relation [6] and the integrated Green
 671 tensor [7] all adjust the single site so that the lattice reproduces the con-
 672 tinuum. For space-filling cubes the adjustment is neither needed nor a free
 673 parameter: the tiling identity fixes the local field exactly, the self term of

674 the collocation closure is the one that makes it cancel, and what is left is a
 675 closed-form error rather than an empirical one. The second-order convergence
 676 reported for the DDA by Yurkin et al. [1] corresponds to the uniform-field
 677 voxel's, and the opposite-signed constants of Eq. (28) explain why a polaris-
 678 ability correction can improve one observable only at the expense of another,
 679 as the package's own form-factor corrections do in reflection and transmis-
 680 sion. Going beyond second order requires a richer internal field, which is the
 681 point of higher-order bases for volume integral equations; the first-moment
 682 voxel is the lowest such basis, and in its spectral form the Galerkin double
 683 average makes the lattice sum converge without Ewald splitting.

684 *The cost of fourth order.* A first-moment voxel carries four times as many
 685 unknowns as a uniform one (36 against 9 in three dimensions). The closed
 686 forms show what that buys. To reach an error ε in reflection, the uniform
 687 voxel needs $kd = \sqrt{8\varepsilon}$ and the first-moment voxel $kd = (720\varepsilon)^{1/4}$; at $\varepsilon = 10^{-6}$
 688 that is $kd = 2.8 \times 10^{-3}$ against 0.16, voxels 58 times larger in each direction.
 689 In a three-dimensional model the voxel count falls by $58^3 \approx 2 \times 10^5$ and the
 690 number of unknowns by about 5×10^4 (these are the Born-order constants;
 691 at the contrast studied the first-moment voxel's measured errors lie within
 692 5% of them, while the uniform voxel's transmission constant grows with the
 693 contrast, so the comparison is if anything conservative). The stratified model
 694 of §9 measures it directly: 347 planes of uniform-field voxels against 8 first-
 695 moment voxels for the same accuracy. Higher moments continue the trend
 696 in one dimension: by Eq. (32) the second-moment voxel reaches $\varepsilon = 10^{-6}$ at
 697 $kd = 0.68$ and the third-moment voxel at $kd = 1.5$, where a voxel is already
 698 a quarter of a wavelength. A three-dimensional voxel of degree p has not

699 been built; with the Legendre polynomials of total degree at most p it would
700 carry 10 ($p = 2$) or 20 ($p = 3$) weights per state component, against 4 for
701 the first-moment voxel. The gain is largest precisely where it matters: when
702 the model varies on scales that are coarse compared with the accuracy the
703 wavefield must have.

704 *Limitations and outlook.* The results concern laterally uniform media, a ho-
705 mogeneous layer and a randomly stratified one, for which the discrete system
706 is exactly a chain of planes; a laterally heterogeneous medium needs the first-
707 moment voxel in a full three-dimensional solver, where the gradient channels
708 couple through the evanescent lattice orders (§8) and the tiling identity holds
709 only to $O(k_{\parallel}h)$. The smoothly graded sphere of §10 shows the uniform-field
710 voxel converging on a laterally heterogeneous body against an exact solution;
711 the fourth order has still to be shown there. A contrast that varies within
712 a cell, which §9 shows is needed beyond second order in a smoothly varying
713 medium, has a further consequence in three dimensions: its linear part is odd
714 about the cell’s centre, so it couples the single site’s uniform-field and first-
715 moment blocks, which parity keeps apart for a homogeneous cube (§6.3). A
716 graded voxel therefore needs at least the first-moment block; it has not been
717 built. The numerical evidence is at moderate, physically typical contrasts
718 (up to 15% in density and moduli) and at $k_S h \leq 0.3$, the range in which
719 the single-site T -matrix is validated; the closed forms are first order in the
720 contrast, and the departures from them at the full contrast are of second or-
721 der. What is established is the *order* of convergence and its leading constant,
722 not the behaviour at strong contrast, where the single site’s own closure may
723 limit the accuracy before the lattice coupling does. The single site’s dynamic

error is in closed form for a sphere only; for a cube, for which no exact solution exists, it is known through the layer (§§7–8). Two extensions follow directly. The first is the three-dimensional solver itself. The second is a stratified *background*, a layered reference medium, as distinct from the stratified contrast of §9. There the lattice coupling then carries the reverberation of the layering, which is not translation invariant in depth, and whether a tiling identity holds for it is the natural next question.

12. Conclusions

A lattice of cubic voxels reproduces a continuous elastic medium in a precise sense. Because cubes tile the plane, the source-cell-averaged coupling between voxels is the continuum’s cell integral exactly, and at normal incidence the lattice kernel is available in closed form with no lattice sum. The single site is derived rather than assumed. The moments of the Green’s tensor over the cube, taken as distributions, give a closed hierarchy whose first-gradient truncation is the uniform-strain closure. That closure is exact for a sphere in the static limit, and it departs from exact scattering by a sphere only by a real $(ka)^2$ term, with the radiation damping exact. For a cube it differs from the sphere only by a cubic anisotropy of the shear response. The collocation closure carries the same self term as the lattice kernel, which cancels, so the discrete system is a collocation of the continuum equation. It converges at second order, with a leading error that is a midpoint-rule error of the internal field, known in closed form for every discretisation, and opposite in sign in reflection and transmission. A voxel that also carries the first moment of its internal field converges at fourth order, at normal incidence (where the

three-dimensional scheme reduces exactly to one dimension) and at oblique
 incidence, with a leading error, $-(kd)^4/720$, that no longer depends on the
 direction of observation. With Legendre moments to degree p it converges at
 order $2p + 2$ with the closed-form constant of Eq. (32), and a single plane of
 such voxels reproduces the thin-layer series term by term through $O(D^{2p+2})$.
 Where the medium varies within a cell, a contrast sampled as constant caps
 every voxel at second order; with the contrast projected to degree r the order
 is $\min(2p+2, 2r+2)$, and the coupling needs nothing beyond the same kernel
 moments to degree $p + r$. The orders survive the interfaces of a randomly
 stratified model, provided voxel boundaries coincide with its interfaces, and
 there the first-moment voxel reaches an accuracy of 10^{-6} with one voxel per
 model layer, where the uniform-field voxel needs about forty. A sphere whose
 contrast falls smoothly to zero at its surface converges against its exact so-
 lution, to 0.061% at $k_{sa} = 0.5$ with twelve voxels across; a sharp sphere's
 error is set by its staircase, which changes with the grid. The results, and
 the evidence for each, are collected in Table 10.

Acknowledgments

This work began thirty years ago in the author's doctoral research at
 the Research School of Earth Sciences of the Australian National University,
 Canberra [9].

Declaration of competing interest

The author declares no known competing financial interests or personal
 relationships that could have appeared to influence the work reported in this

Table 10: The results of the paper, and the evidence for each.

Result	Evidence
Continuous layer, exact and to $O(D^k)$	reference to 2.5×10^{-8} ; series order k
Thin-layer series, Eq. (5)	each coefficient symbolically, to D^4
Point-kernel local term survives $d \rightarrow 0$	$11 \times$ the continuum term at $m = 1$
Averaged coupling between planes = continuum $\times$ sinc	2.8×10^{-16} , 81 entries
Tiling identity, Eq. (11)	21/21 components to 4×10^{-9}
Exact kernel, Eq. (12)	replaces a 9.4×10^{-4} static error
Moments as distributions, Eq. (19)	sum rule (20) exact; M, Q equal an independent archive, 81/81
Channels (21)–(23)	a production code by another route, 2×10^{-15}
Cube vs sphere, Eq. (24)	symbolic; and numerically at five ν , 4×10^{-16}
Ball closure = exact Mie, static	2×10^{-10} ; cube's bulk and shear average 10^{-8}
Ball closure vs Mie, dynamic, Eq. (25)	symbolic; numerical Mie 5×10^{-4} ; $(ka)^3$ terms exact
Self term cancels, Eq. (27)	4×10^{-16}
Package T = closure statically	2×10^{-5} at $k_S h = 0.01$
Second-order convergence	order 2.000, three frequencies
Closed-form error, Eq. (28)	$n = 1$ –16, to 10^{-4} of the error
First-moment voxel, Eq. (30)	order 4.00, three frequencies
First-moment voxel, oblique, Eq. (33)	order 4, P and S, 20° and 40°

771 paper.

772 **Data availability**

773 The Mathematica notebooks and Python scripts that derive and check ev-
774 ery result, and the package they test, are archived at [https://doi.org/10.](https://doi.org/10.5281/zenodo.23003241)
775 [5281/zenodo.23003241](https://doi.org/10.5281/zenodo.23003241) and developed at [https://github.com/tmnestor/](https://github.com/tmnestor/MultipleScatteringCalculations)
776 [MultipleScatteringCalculations](https://github.com/tmnestor/MultipleScatteringCalculations).

777 **References**

- 778 [1] M. A. Yurkin, V. P. Maltsev, A. G. Hoekstra, Convergence of the dis-
779 crete dipole approximation. I. Theoretical analysis, J. Opt. Soc. Am. A
780 23 (2006) 2578–2591.
- 781 [2] L. L. Foldy, The multiple scattering of waves. I. General theory of
782 isotropic scattering by randomly distributed scatterers, Phys. Rev. 67
783 (1945) 107–119.
- 784 [3] M. Lax, Multiple scattering of waves, Rev. Mod. Phys. 23 (1951) 287–
785 310.
- 786 [4] E. M. Purcell, C. R. Pennypacker, Scattering and absorption of light by
787 nonspherical dielectric grains, Astrophys. J. 186 (1973) 705–714.
- 788 [5] B. T. Draine, P. J. Flatau, Discrete-dipole approximation for scattering
789 calculations, J. Opt. Soc. Am. A 11 (1994) 1491–1499.

- [6] B. T. Draine, J. Goodman, Beyond Clausius–Mossotti: wave propagation on a polarizable point lattice and the discrete dipole approximation, *Astrophys. J.* 405 (1993) 685–697.
- [7] P. C. Chaumet, A. Sentenac, A. Rahmani, Coupled dipole method for scatterers with large permittivity, *Phys. Rev. E* 70 (2004) 036606.
- [8] J. D. Eshelby, The determination of the elastic field of an ellipsoidal inclusion, and related problems, *Proc. R. Soc. Lond. A* 241 (1957) 376–396.
- [9] T. M. Nestor, *Seismic Wave Propagation in Complex Media: Summing the Multiple Scattering Series*, Ph.D. thesis, Research School of Earth Sciences, Australian National University, Canberra, ACT, Australia, 1996.
- [10] E. M. Stein, G. Weiss, *Introduction to Fourier Analysis on Euclidean Spaces*, Princeton University Press, Princeton, NJ, 1971.
- [11] C. M. Linton, Lattice sums for the Helmholtz equation, *SIAM Rev.* 52 (2010) 630–674. doi:10.1137/09075130X.
- [12] J. D. Eshelby, The elastic field outside an ellipsoidal inclusion, *Proc. R. Soc. Lond. A* 252 (1959) 561–569. doi:10.1098/rspa.1959.0173.
- [13] T. Mura, *Micromechanics of Defects in Solids*, 2nd ed., Martinus Nijhoff, Dordrecht, 1987. doi:10.1007/978-94-009-3489-4.
- [14] Y. Mikata, S. Nemat-Nasser, Elastic field due to a dynamically

- transforming spherical inclusion, J. Appl. Mech. 57 (1990) 845–849.
doi:10.1115/1.2897650.
- [15] T. M. Michelitsch, H. Gao, V. M. Levin, Dynamic Eshelby tensor and potentials for ellipsoidal inclusions, Proc. R. Soc. Lond. A 459 (2003) 863–890. doi:10.1098/rspa.2002.1054.
- [16] Y. P. Chiu, On the stress field due to initial strains in a cuboid surrounded by an infinite elastic space, J. Appl. Mech. 44 (1977) 587–590. doi:10.1115/1.3424140.
- [17] J. K. Lee, W. C. Johnson, Calculation of the elastic strain field of a cuboidal precipitate in an anisotropic matrix, Phys. Status Solidi A 46 (1978) 267–272. doi:10.1002/pssa.2210460134.
- [18] G. J. Rodin, Eshelby’s inclusion problem for polygons and polyhedra, J. Mech. Phys. Solids 44 (1996) 1977–1995. doi:10.1016/S0022-5096(96)00066-X.
- [19] H. Nozaki, M. Taya, Elastic fields in a polyhedral inclusion with uniform eigenstrains and related problems, J. Appl. Mech. 68 (2001) 441–452. doi:10.1115/1.1362670.
- [20] A. K. Mal, L. Knopoff, Elastic wave velocities in two-component systems, J. Inst. Math. Appl. 3 (1967) 376–387. doi:10.1093/imamat/3.4.376.
- [21] J. E. Gubernatis, E. Domany, J. A. Krumhansl, Formal aspects of the theory of the scattering of ultrasound by flaws in elastic materials, J. Appl. Phys. 48 (1977) 2804–2811. doi:10.1063/1.324141.

- [22] S. K. Kanaun, V. M. Levin, Self-Consistent Methods for Composites, Vol. 2: Wave Propagation in Heterogeneous Materials, Solid Mechanics and Its Applications, Springer, Dordrecht, 2008. doi:10.1007/978-1-4020-6968-0.
- [23] R.-S. Wu, K. Aki, Scattering characteristics of elastic waves by an elastic heterogeneity, *Geophysics* 50 (1985) 582–595. doi:10.1190/1.1441934.
- [24] P. C. Waterman, Matrix theory of elastic wave scattering, *J. Acoust. Soc. Am.* 60 (1976) 567–580. doi:10.1121/1.381130.
- [25] V. D. Kupradze, Dynamical Problems in Elasticity, volume 3 of *Progress in Solid Mechanics*, North-Holland, Amsterdam, 1963.
- [26] K. Aki, P. G. Richards, Quantitative Seismology, 2nd ed., University Science Books, Sausalito, CA, 2002.
- [27] B. L. N. Kennett, Seismic Wave Propagation in Stratified Media, Cambridge University Press, Cambridge, 1983.
- [28] C. F. Ying, R. Truell, Scattering of a plane longitudinal wave by a spherical obstacle in an isotropically elastic solid, *J. Appl. Phys.* 27 (1956) 1086–1097. doi:10.1063/1.1722545.
- [29] Y.-H. Pao, C.-C. Mow, Diffraction of Elastic Waves and Dynamic Stress Concentrations, Crane Russak, New York, 1973.
- [30] H. Takeuchi, M. Saito, Seismic surface waves, in: B. A. Bolt (Ed.), *Methods in Computational Physics*, volume 11, Academic Press, New York, 1972, pp. 217–295.

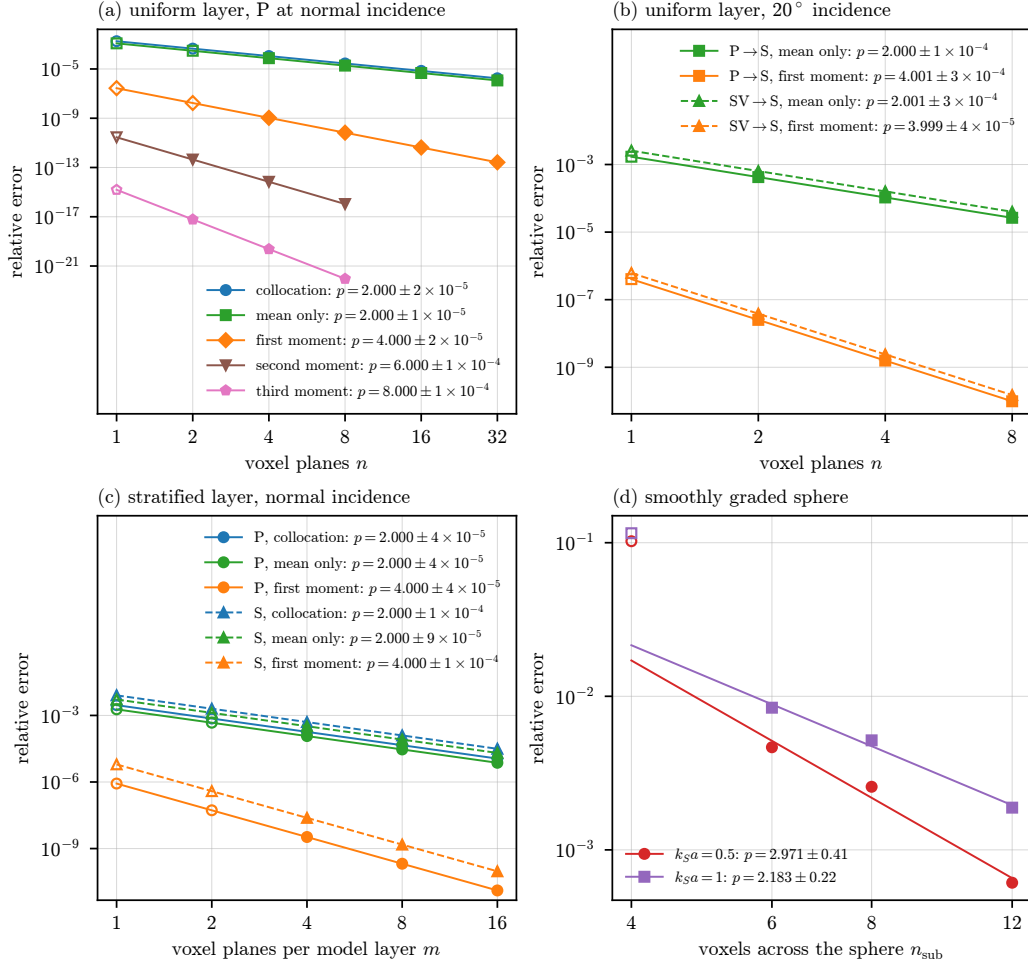


Figure 4: Convergence of every test, with least-squares regression lines. Each line fits $\log e = c - p \log n$ to the filled points; open points are pre-asymptotic and are not fitted. The legend gives the order p and its standard error. Colour marks the scheme: collocation (§6), the mean-only voxel and the voxels carrying Legendre moments to degree $p = 1, 2$ and 3 (Eqs. (30) and (33)). (a) The uniform layer, P reflection at normal incidence ($D = 2$ m, $\omega = 300$ rad s $^{-1}$, the contrast of §7), fitted from $n = 4$; the second- and third-moment voxels, computed in 40-digit arithmetic, fitted from $n = 2$. (b) The uniform layer at 20°, fitted from $n = 2$: the S wave converted from incident P (solid) and reflected from incident SV (dashed). (c) The stratified layer of §9, reflection at normal incidence for P (solid) and S (dashed), against the number m of voxel planes per model layer, fitted from $m = 4$. (d) The smoothly graded sphere of §10 against its exact solution, for n_{sub} voxels across, fitted from $n_{\text{sub}} = 6$ (at 4 a voxel is as wide as the graded shell).

Alt text: Four log-log panels of relative error against refinement, each showing data points and fitted straight lines with the fitted order in the legend. (a) Uniform layer at normal incidence, 1 to 32 voxel planes: the collocation and mean-only lines have order 2.000, the first-moment line order 4.000, falling from about 3×10^{-7} to 2×10^{-13} ; the